

Solutions to JEE Main - 1 | JEE-2022

PHYSICS

SECTION-1

1.(D) $N = F - mg \cos \theta \quad \dots (i)$

$f - mg \sin \theta = 0 \quad \dots (ii)$

For limiting case, force applied is minimum.

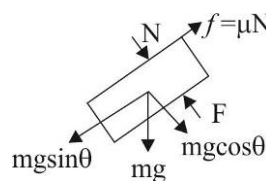
$\mu(F - mg \cos \theta) = mg \sin \theta$

$$F = \frac{mg \sin \theta}{\mu} + mg \cos \theta$$

$$F = mg \left(\frac{\sin \theta}{\mu} + \cos \theta \right)$$

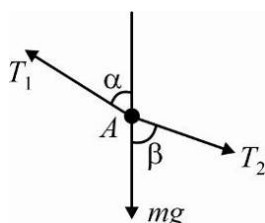
$$F = 100 \left(\frac{\sin 37^\circ}{1/2} + \cos 37^\circ \right)$$

$$F = 100 \left(2 \times \frac{3}{5} + \frac{4}{5} \right) = 200 N$$

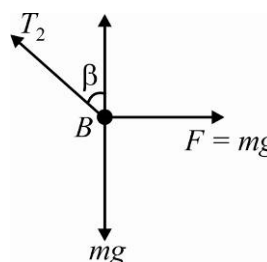


2.(D) $T_2 \cos \beta = mg \quad \dots (i)$

$T_2 \sin \beta = F = mg \quad \dots (ii)$

From equations (i) and (ii), $\beta = 45^\circ$


$T_2 \cos \beta + mg = T_1 \cos \alpha \quad \dots (iii)$



$T_2 \sin \beta = T_1 \sin \alpha \quad \dots (iv)$

Dividing equation (iv) and (iii)

$$\tan \alpha = 1/2 \Rightarrow \sin \alpha = 1/\sqrt{5}$$

From equation (iv), $T_2 \frac{1}{\sqrt{2}} = T_1 \frac{1}{\sqrt{5}} \Rightarrow \sqrt{5}T_2 = \sqrt{2}T_1$

3.(A) Let that vector be $\vec{C}$. Then $\vec{C} = C \hat{C} = b\hat{a} \Rightarrow \vec{C} = \frac{b\vec{a}}{a} = \frac{5}{\sqrt{2}}(\hat{i} - \hat{j})$

4.(C) $s_{\text{man, bus}} = u_{\text{man, bus}} t + \frac{1}{2} a_{\text{man, bus}} t^2$

$16 = 8t - \frac{1}{2} \times 2t^2 \quad (\text{equation relative to bus})$

$t = 4s$

5.(C)

6.(B) $\tan 30^\circ = \frac{u \sin \theta - g \times 2}{u \cos \theta}$

and $\frac{1}{2} T_f = 3 \Rightarrow \frac{u \sin \theta}{g} = 3 \Rightarrow u \sin \theta = 3g$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{3g-2g}{u \cos \theta} \Rightarrow u \cos \theta = \sqrt{3}g$$

$$\Rightarrow \tan \theta = \sqrt{3} \Rightarrow \theta = 60^\circ, u = 2\sqrt{3}g = 20\sqrt{3} \text{ m/s}$$

$$\begin{aligned} 7.(B) \quad y = ax^2, \quad \frac{dy}{dt} = a(2x) \frac{dx}{dt} = 2acx, \quad \frac{d^2y}{dt^2} = 2ac \frac{dx}{dt} = 2ac^2 \\ a_y = 2ac^2, \quad a_x = 0, \quad \vec{a} = a_x \hat{i} + a_y \hat{j} \quad \therefore \quad \vec{a} = 2ac^2 \hat{j} \quad \therefore \quad (B) \end{aligned}$$

8.(B)

9.(A) The velocity v acquired by the parachutist after 10 s

$$v = u + gt = 0 + 10 \times 10 = 100 \text{ m/s}$$

$$\text{Then, } s_1 = ut + \frac{1}{2}gt^2 = 0 + \frac{1}{2} \times 10 \times 10^2 = 500 \text{ m}$$

The distance travelled by the parachutist under retardation, $s_2 = 2495 - 500 = 1995 \text{ m}$

Let v_g be his velocity on reaching the ground.

$$\text{Then } v_g^2 - v^2 = 2as_2 \text{ or } v_g^2 - (100)^2 = 2 \times (-2.5) \times 1995 \text{ or } v_g = 5 \text{ m/s}$$

10.(A) Particle comes to rest when slope of $x-t$ curve becomes zero (6 times)

$$11.(C) \text{ Average speed} = \frac{\text{distance}}{\text{time}} = \frac{\text{area under } v-t \text{ graph}}{\text{time}} = \frac{1}{1} \left(\frac{1}{2} \times (1+2) \times 1 \right) = 1.5 \text{ m/s}$$

12.(A) From the graph acceleration $\propto t$

$\therefore$ Velocity $\propto t^2$

$\therefore$ Graph would be a parabola opening upwards for time t_1 & opening downwards thereafter.

$$13.(B) \quad a = \frac{v dv}{ds} \Rightarrow \int a ds = \int_{v_i}^{v_f} v dv$$

$$\Rightarrow \text{area under } a-s \text{ curve} = \frac{v_f^2 - v_i^2}{2} = 0.4$$

$$v_f^2 = (0.8)^2 + 2 \times 0.4; \quad v_f = 1.2 \text{ m/s}$$

$$14.(A) \quad v = (3x^2 + x) \text{ m/s}$$

$$a = v \frac{dv}{dx} \left(v_{x=2} = 14 \text{ m/s}; \left. \frac{dv}{dx} \right|_{x=2} = 13 \text{ m/s} \right) = 14 \times 13 = 182 \text{ m/s}$$

15.(A) Displacement perpendicular to inclined plane,

$$0 = v \sin \theta t - \frac{1}{2} g \cos \theta t^2 \Rightarrow t = \frac{2v \sin \theta}{g \cos \theta} = \frac{2 \times 5 \times 3/5}{10 \times 4/5} = \frac{3}{4}$$

$$16.(D) \quad \frac{dx}{dt} = ky; \quad \frac{dy}{dt} = kx$$

Dividing both equation & integrating, we get $\frac{dy}{dx} = \frac{x}{y} \Rightarrow \int y dy = \int x dx$

$$y^2 = x^2 + \text{constant}$$

$$17.(B) \quad a_t = 6t \, m/s^2 \quad \Rightarrow \quad R\alpha = 6t; \quad \frac{d\omega}{dt} = \frac{6t}{R} \quad \Rightarrow \quad \omega = \frac{3}{4} \pi t^2$$

$$\frac{d\theta}{dt} = \frac{3}{4} \pi t^2 \quad \Rightarrow \quad \int_0^{2\pi} d\theta = \frac{3}{4} \pi \int_0^t t^2 dt \quad \Rightarrow \quad t = 2s$$

$$\therefore \quad a_t = 12 \, m/s^2 \quad \text{and} \quad a_r = \omega^2 R = 36\pi; \quad a = \sqrt{a_t^2 + a_r^2} = \sqrt{(12)^2 + (36\pi)^2} = 12\sqrt{1+9\pi^2}$$

$$18.(B) \quad V_y = u_y + a_y t; \quad 0 = 5 - g \cos 30^\circ t \quad \Rightarrow \quad t = \frac{1}{\sqrt{3}} \, \text{sec}.$$

$$19.(C) \quad \text{Let applied force is } F_4 = F_x \hat{i} + F_y \hat{j}$$

To prevent the particle P from moving, net force on the particle will be zero.

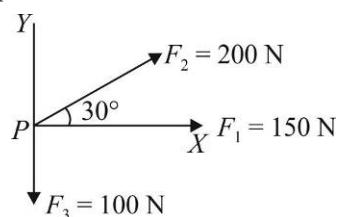
$$\therefore \quad \sum F_x = 0$$

$$\text{or} \quad \sum F_x = F_x + 150 + 200 \cos 30^\circ = 0$$

$$\therefore \quad F_x = -150 - 100\sqrt{3} = -(150 + 170) = -320 \, \text{N}$$

$$\text{and} \quad F_y = 0$$

It means $\vec{F}_4$ is directed opposite to $\vec{F}_1$.



$$20.(C) \quad \vec{a} \times \vec{b} \text{ is perpendicular to the plane of } \vec{a} \text{ and } \vec{b}$$

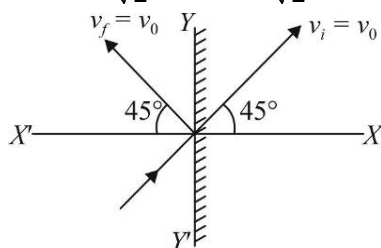
unit vector along $\vec{a} \times \vec{b}$,

$$\begin{aligned} \therefore \quad \hat{n} &= \frac{\vec{a} \times \vec{b}}{|\vec{a}| |\vec{b}|} = \frac{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & 1 \\ 1 & 2 & 2 \end{vmatrix}}{(\sqrt{4+4+1}\sqrt{1+4+4})} \\ &= \frac{\hat{i}(-4-2) - \hat{j}(4-1) + \hat{k}(4+2)}{9} \\ &= -\frac{6}{9}\hat{i} - \frac{3}{9}\hat{j} + \frac{6}{9}\hat{k} = -\frac{2}{3}\hat{i} - \frac{1}{3}\hat{j} + \frac{2}{3}\hat{k} \end{aligned}$$

SECTION-2

$$1.(2) \quad \text{Initial velocity is } v = v_0 \cos 45^\circ \hat{i} + v_0 \sin 45^\circ \hat{j}$$

$$= 5\sqrt{2} \times \frac{1}{\sqrt{2}} \hat{i} + 5\sqrt{2} \times \frac{1}{\sqrt{2}} \hat{j} = 5\hat{i} + 5\hat{j}$$



The final velocity

$$\begin{aligned} v_f &= -v_0 \cos 45^\circ \hat{i} + v_0 \sin 45^\circ \hat{j} \\ &= -5\sqrt{2} \times \frac{1}{\sqrt{2}} \hat{i} + 5\sqrt{2} \times \frac{1}{\sqrt{2}} \hat{j} = -5\hat{i} + 5\hat{j} \end{aligned}$$

$\therefore$ The change in velocity is

$$\begin{aligned} \Delta v &= v_f - v_i \\ &= (-5\hat{i} + 5\hat{j}) - (5\hat{i} + 5\hat{j}) = -10\hat{i} \quad \therefore \quad |\Delta v| = 10 \text{ ms}^{-1} \end{aligned}$$

2.(5) $u = 7 \text{ m/s}$ and $a = 4 \text{ m/s}^2$

Distance traveled in n^{th} second $= u + \frac{a}{2}(2n-1)$

$\therefore$ Distance traveled in 5^{th} second $= 7 + \frac{4}{2}[2(5)-1] = 25\text{m} = 5 \times 5\text{m} \Rightarrow x = 5$

3.(9) Given that $u = 0$ (the particle starts from rest)

At any time $t : v = kt = 2t$. $a = \frac{dv}{dt} = 2 \text{ m/s}^2$ (constant acceleration)

Now $s = ut + \frac{1}{2}at^2 = 0 \times 3 + \frac{1}{2} \times 2 \times (3)^2 = 9\text{m}$

4.(2) $\Delta t = \frac{2 \times 10}{10} = 2\text{sec}$

5.(1) If $\vec{a}$ and $\vec{b}$ are perpendicular, $\vec{a} \cdot \vec{b} = 0$
or $2x - 3 + 1 = 0 \quad \therefore \quad x = 1$

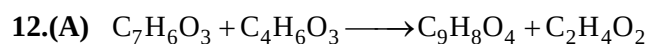
Chemistry

SECTION-1

- 1.(C) According to $(n + \ell)$ rule, correct order of energy will be :
 $3s < 3d < 4p$
 $4s < 3d < 4p$
 $3d < 4p < 5s$
 $4p < 5s < 4d$
- 2.(B) Configuration of Cu^{2+} will be :
 $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^9$
- 3.(C) Order of I.E. : $\text{C} > \text{B} > \text{Si} > \text{Al}$
- 4.(C) Order of electronegativity : $\text{O} > \text{N} > \text{Br} > \text{S}$
- 5.(A) Order of atomic radius :
 $\text{Na} > \text{Li} > \text{Be} > \text{B}$
- 6.(D) There is smallest difference in first I.E. in Fe, Co, Ni because electron is removed from 4s in all three.
- 7.(C) Let mass of compound to be 100 gm.
- | | |
|----------------------------|---------------------|
| Re | Cl |
| 63.6 gm | 36.4 gm |
| moles = $\frac{63.6}{186}$ | $\frac{36.4}{35.5}$ |
| = 0.342 | = 1.025 |
| ≈ 1 | 3 |
- Empirical formula of compound will be : ReCl_3
- 8.(D) $\text{SnO}_2 + 2\text{H}_2 \rightarrow \text{Sn} + 2\text{H}_2\text{O}$
- | | |
|-----------------------------|--|
| 2gm | $\downarrow$ |
| $\frac{2}{151} \text{ mol}$ | $\frac{4}{151} \text{ mol}$ |
| $\downarrow$ | |
| | $\frac{4}{151} \times 22.4 \text{ lit.}$ |
| | = 0.593 lit. |
- 9.(B) $\text{P} \longrightarrow \text{P}^{3-}$
- | | | |
|----------------------------------|----|----|
| Total electrons $\Rightarrow$ | 15 | 18 |
| Unpaired electrons $\Rightarrow$ | 3 | 0 |
- 10.(D) Correct order of metallic character will be :
 $\text{Si} > \text{P} > \text{S}$
 $\text{As} > \text{P} > \text{N}$
 $\text{Br} < \text{Se} < \text{As}$

11.(B) $E = -13.6 \frac{Z^2}{n^2} \text{ eV/atom}$

On increasing value of n , E increases but energy difference between successive energy levels decreases.



1gm

$\Downarrow$

$$\frac{1}{138.12} \text{ mol} \longrightarrow \frac{1}{138.12} \text{ mol}$$

$$= \frac{1}{138.12} \times 180.15 = 1.304 \text{ gm}$$

$$\% \text{ yield} = \frac{0.85}{1.304} \times 100 = 65.18\%$$

13.(C) Atomic radius

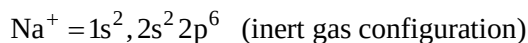
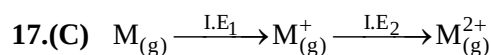
14.(B) 2p will be next higher energy subshell.

For 2p, $n = 2$

$$\ell = 1$$

15.(D) Order of I.E. $\Rightarrow C < O < N$

16.(B) Al and Ga have mostly similar chemical properties.



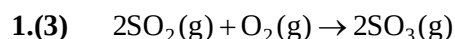
So removal of one more e^- from Na^+ will be difficult Hence its I.E. will be higher.

18.(A) Transition from $n = 3 \rightarrow n = 1$, will produce a photon with highest energy among given options.

19.(D) The emission spectrum of hydrogen atom in visible region consists of a series of lines that are closer at higher energies.

20.(B) XZ plane is Nodal plane in p_y orbital.

SECTION-2



Under identical conditions of T and P, equal volume of all gases contains equal number of molecules

So 2V of SO_2 requires V volume of O_2 to give 2 V volume of SO_3

So 3 dm³ of SO_2 requires 1.5 dm³ of O_2 to give 3 dm³ of SO_3

Volume of O_2 given = 2 dm³

So O_2 is present in excess and SO_2 is the limiting reagent

Volume of SO_3 formed = 3dm³



1 mol of Fe_2O_3 requires 3 mol of C

So 6 mol of Fe_2O_3 will require 18 mol of C

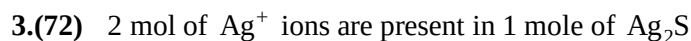
But only 9 mol of C is present.

So the limiting reagent is carbon

3 mol of C gives 2 mole of Fe

So 9 mol of C will give 6 mol of Fe

$\therefore$ theoretical yield of Fe = 6 mol

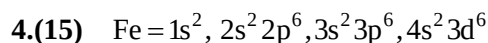


$$\text{Moles of } \text{Ag}_2\text{S} \text{ formed} = \frac{g}{M_0} = \frac{1.24}{(108 \times 2) + 32} = \frac{1.24}{248} = 0.005$$

So mol of Ag^+ ions that must have been present = 0.01 mol

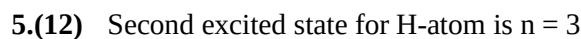
Mass of 0.01 mol Ag^+ ions = $0.01 \times 108 = 1.08$ g

$$\begin{aligned} \% \text{ of silver in the ore} &= \frac{\text{Mass of Silver}}{\text{Mass of Ore}} \times 100 \\ &= \frac{1.08}{1.5} \times 100 = 72\% \end{aligned}$$



Number of orbitals which contain one or more electrons

$$1 + 1 + 3 + 1 + 3 + 1 + 5 = 15$$



For mono-electronic species energy depends only on n

So energy of $3s = 3p = 3d$

Number of degenerate orbitals = 9

Second excited state for H^- ion = 2 p

For polyelectronic species, energy increases in the order : $1s < 2s < 2p$

Number of degenerate orbitals = 3

Total degenerate orbitals = $9 + 3 = 12$

Mathematics

SECTION-1

$$1.(B) \quad \frac{2 \tan \frac{t}{2} + 1 - \tan^2 \frac{t}{2}}{1 + \tan^2 \frac{t}{2}} = \frac{1}{5}$$

$$\tan\left(\frac{t}{2}\right) = \lambda$$

$$10\lambda + 5 - 5\lambda^2 = 1 + \lambda^2$$

$$6\lambda^2 - 10\lambda - 4 = 0$$

$$6\lambda(\lambda - 2) + 2(\lambda - 2) = 0$$

$$\lambda = 2, \quad \lambda = -1/3$$

$$\tan \frac{t}{2} = -\frac{1}{3}$$

$$\text{As } \pi < t < 2\pi$$

$$2.(B) \quad \cos A = -\frac{3}{5} \quad (A \text{ is angle of } \Delta \text{ hence in 2nd quadrant})$$

$$9x^2 + 27x + 20 = 0$$

$$9x^2 + 15x + 12x + 20 = 0$$

$$3x(3x + 5) + 4(3x + 5) = 0$$

$$x = -\frac{4}{3}, \quad x = -\frac{5}{3} \quad \sec A$$

$$\tan A = -\frac{4}{3}$$

$$3.(C) \quad \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$

$$\text{If } \alpha + \beta + \gamma = 0$$

$$2x^3 - x + 1 = 0 \quad (\alpha, \beta, \gamma) \quad (\alpha + \beta + \gamma = 0)$$

$$\alpha\beta\gamma = -\frac{1}{2}$$

$$\alpha^3 + \beta^3 + \gamma^3 = -\frac{3}{2}$$

$$4.(C) \quad xy = 2 \sin t \cos t$$

$$\frac{x}{y} = \frac{\sin t}{\cos t}$$

$$x^2 = 2 \sin^2 t, \quad y^2 = 2 \cos^2 t$$

$$x^2 + y^2 = 2$$

$$5.(C) \quad \frac{\sin \theta + 2 \sin \theta \cos \theta}{\cos \theta + 2 \cos^2 \theta} = \tan \theta$$

$$\begin{aligned}
 6.(C) \quad & 16(\cos 2A - \cos 3A) \\
 & 16(2\cos^2 A - 1 - 4\cos^3 A + 3\cos A) \\
 & 16\left(2 \times \frac{9}{16} - 1 - 4 \times \frac{27}{64} + \frac{3 \times 3}{4}\right) \\
 & 18 - 16 - 27 + 36 \\
 & 38 - 27 = 11
 \end{aligned}$$

$$\begin{aligned}
 7.(A) \quad & 2\alpha + 2\beta + \alpha = 20 \\
 & \alpha = -8 \\
 & 64 - 48 + \lambda = 0 \Rightarrow \lambda = -16
 \end{aligned}$$

$$\begin{aligned}
 8.(B) \quad & x^2 + mx + n = 0 \quad (\alpha, \beta) \\
 & \alpha + \beta = -m, \quad \alpha\beta = n \\
 & x^2 + px + q = 0 \quad (\alpha^3, \beta^3) \\
 & \alpha^3 + \beta^3 = -p \Rightarrow -p = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) \\
 & -p = (-m)^3 - 3n(-m) \\
 & p = m^3 - 3mn
 \end{aligned}$$

$$\begin{aligned}
 9.(B) \quad & a + a + 4d = 26 \Rightarrow a + 2d = 13 \Rightarrow a = 13 - 2d \\
 & (a + d)(a + 3d) = 160 \\
 & (13 - d)(13 + d) = 160 \\
 & d^2 - 169 = -160 \\
 & d^2 = 9 \Rightarrow d = 3 \text{ or } -3 \\
 & a = 7 \text{ or } 19 \\
 & s_6 = \frac{6}{2}(14 + 15) \text{ or } \frac{6}{2}(38 - 15) \\
 & = 3 \times 29 \text{ or } 3 \times 23 \\
 & 87 \text{ or } 69
 \end{aligned}$$

$$\begin{aligned}
 10.(B) \quad & 2 - 2\sin^2 \theta + 3\sin \theta = 0 \\
 & 2\sin^2 \theta - 3\sin \theta - 2 = 0 \\
 & 2t^2 - 3t - 2 = 0 \\
 & 2t^2 - 4t + t - 2 = 0 \\
 & 2t(t - 2) + 1(t - 2) = 0 \\
 & t = -\frac{1}{2} \sin \theta = -\frac{1}{2} \\
 & \theta = -\frac{\pi}{6}, -\frac{5\pi}{6}, \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6} \\
 & \text{Sum} = 2\pi
 \end{aligned}$$

11.(D) a, b, c, d – GP

$$a, ar, ar^2, ar^3$$

$$a + ar^3 = 112$$

$$ar + ar^2 = 48$$

$$\frac{1+r^3}{r(1+r)} = \frac{112}{48} = \frac{7}{3}$$

$$\frac{1+r^2-r}{r} = \frac{7}{3}$$

$$3r^2 - 3r + 3 = 7r$$

$$3r^2 - 10r + 3 = 0$$

$$r = 3$$

$$a(28) = 112$$

$$a = 4$$

$$s_6 = \frac{4(3^6 - 1)}{(3 - 1)} = 2(728)$$

$$= 1456$$

12.(A) $s = \frac{1}{2^1} + \frac{2}{2^2} + \frac{3}{2^3} + \dots + \frac{20}{2^{20}}$

$$\frac{s}{2} = \frac{1}{2^2} + \frac{2}{2^3} + \dots + \frac{19}{2^{20}} + \frac{20}{2^{21}}$$

$$\frac{s}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots + \frac{1}{2^{20}} - \frac{2}{2^{21}}$$

$$\frac{s}{2} = \frac{\frac{1}{2} \left(1 - \left(\frac{1}{2} \right)^{20} \right)}{\frac{1}{2}} = \frac{20}{2^{21}}$$

$$s = 2 - \frac{1}{2^{19} - \frac{10}{2^{19}}} \Rightarrow 2 - \frac{11}{2^{19}}$$

13.(B) $\frac{\sin(2\alpha + \beta)}{\sin \beta} = 3$

Apply C & D

$$\frac{\sin(2\alpha + \beta) + \sin \beta}{\sin(2\alpha + \beta) - \sin \beta} = \frac{4}{2}$$

$$\frac{2 \sin(\alpha + \beta) \cos \alpha}{2 \cos(\alpha + \beta) \sin \alpha} = 2$$

$$\frac{\tan(\alpha + \beta)}{\tan \alpha} = 2$$

14.(A) $\frac{1^3}{1} + \frac{1^3 + 2^3}{1+2} + \dots + \frac{1^3 + 2^3 + \dots + 15^3}{1+2+3+\dots+15} - \frac{1}{2}(1+2+\dots+15) = \frac{15 \times 16}{4}$

$$T_k = \frac{\frac{k^2(k+1)^2}{4}}{\frac{k(k+1)}{2}} = \frac{(k+1)k}{2}$$

$$s_k = \left(\frac{k^2}{2} + \frac{k}{2} \right)$$

$$\frac{(15)(16)(31)}{12} + \frac{15 \times 16}{4} - \frac{15 \times 16}{4}$$

$$s_k m = 620$$

15.(C) $x^2 + mx + 1 < 3x^2 + 3x + 3$

$$2x^2 + (3-m)x + 2 > 0$$

$$D < 0$$

$$(3-m)^2 - 16 < 0$$

$$-4 < m < -3 < 4$$

$$-1 < m < 7$$

$$\frac{x^2 + mx + 1}{x^2 + n + 1} > -3$$

$$4x^2 + (3+m)x + 4 > 0$$

$$D < 0$$

$$(m+3)^2 - 64 < 0 \Rightarrow m \in (-11, 5)$$

16.(C) a, b, c - G.P

$$a \quad ar \quad ar^2$$

$$2b + 2c = a + b + c + a$$

$$b + c = 2a$$

$$ar + ar^2 = 2a$$

$$r^2 + r - 2 = 0$$

$$(r+2)(r-1) = 0$$

$$r = -2$$

17.(A) $\frac{1}{d} \left(\frac{d}{a_1 a_2} + \frac{d}{a_2 a_3} + \frac{d}{a_3 a_4} + \dots \right)$

$$\frac{1}{d} \left(\frac{1}{a_1} - \frac{1}{a_2} + \frac{1}{a_2} - \frac{1}{a_3} + \dots \right)$$

$$\frac{1}{a_1 d}$$

18.(D) HCF $(91, n) > 1$

$$\Rightarrow n \text{ is multiple of } 7 \text{ or } 13$$

$$105, 112, 196 \quad \frac{14}{2} (105 + 116) = 2107$$

$$\text{Or } 104, 127, \dots 195 = \frac{8}{2} (104 + 195)$$

$$\text{But } 182 \text{ common to } 1196 \text{ both}$$

$$\text{Sum} = 3121$$

19.(D) a, b, c

$$a \quad ar \quad ar^2$$

$$3a, 7b, 15c \text{ are A.P.}$$

$$14ar = 3a + 15ar^2$$

$$15r^2 - 14r + 3 = 0$$

$$15r^2 - 9r - 5r + 3 = 0$$

$$3r(5r - 3) - (5r - 3) = 0$$

$$r = \frac{3}{5}, r = +\frac{1}{3}$$

$$3a, \frac{7a}{3}, \frac{5a}{3}, \frac{3a}{3}$$

20.(B) $x^2 + (a-b)x + 1 - a - b = 0$ has real & distinct root

$$D > 0$$

$$(a-b)^2 - 4(1-a-b) > 0$$

$$b^2 - 2ab + 4b + a^2 + 4a - 4 > 0 \quad \forall b \in R$$

$$b^2 + 2b(2-a) + a^2 + 4a - 4 > 0 \quad \forall b \in R$$

$$D < 0$$

$$4(2-a)^2 - 4(a^2 + 4a - 4) < 0$$

$$-16a + 16 - 16a + 16 < 0$$

$$32a > 32 \Rightarrow a > 1$$

SECTION-2

1.(7) $\sec x = \frac{1}{1 - \cos x}$

$$\Rightarrow \sec x = 2$$

$$\cos x = \frac{1}{2} \quad (0, 2\pi) - [\pi]$$

$$x = \frac{\pi}{3}, \quad 2\pi - \frac{\pi}{3}$$

$$x_2 - x_1 = \frac{4\pi}{3}$$

2.(3) $\cos x \cos \frac{\pi}{3} - \sin x \sin \left(\frac{\pi}{3}\right) + \cos x$

$$E = \frac{3\cos x}{2} - \frac{\sqrt{3}}{2} \sin x$$

$$E_{\text{Max}} = \sqrt{\frac{9}{4} + \frac{3}{4}} = \sqrt{\frac{12}{4}} = \sqrt{3}$$

3.(11) $Ax^2 - 4x + 1 = 0 \quad (\alpha, \gamma)$

$$x^2 - 4x + A = 0 \quad (1/\alpha, 1/\gamma)$$

$$2a + 2d = 4 \Rightarrow a + d = 2$$

$$Bx^2 - 6x + 1 = 0 \quad (\beta, \delta)$$

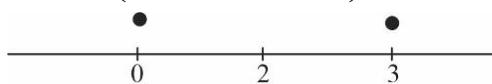
$$x^2 - 6x + B = 0 \quad (1/\beta, 1/\gamma)$$

$$\alpha, \beta, \gamma, \delta - \text{H.P.}$$

$$\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}, \frac{1}{\delta} - \text{A.P.}$$

$$B = 8, A = 3, A + B = 7$$

4.(23) $f(x) = \left(x^2 - \frac{2c}{c-5}x + \frac{c-4}{c-5} \right)$



$$f(0) > 0, \quad f(2) < 0, \quad f(3) > 0$$

$$\frac{c-4}{c-5} > 0, \quad c \in (-\infty, 4) \cup (5, \infty)$$

$$4 - \frac{4c}{c-5} + \frac{c-4}{c-5} < 0 \Rightarrow \frac{4c-20-4c+c-4}{c-5} < 0 \quad \frac{c-24}{c-5} < 0 \quad c \in (5, 24)$$

$$9 - \frac{6c}{c-5} + \frac{c-4}{c-5} > 0 \Rightarrow \frac{9c-45-6c+c-4}{c-5} > 0$$

$$\frac{4c-49}{c-5} > 0$$

$$c \in (-\infty, 5) \cup \left(\frac{49}{4}, \infty \right) \text{ find } c \in \left(\frac{49}{4}, 24 \right) \quad c \text{ largest} = 23$$

5.(25) $3x^2 - 10x - 25 = 0$ ($\tan A, \tan B$)

$$\tan A + \tan B = \frac{10}{3}, \quad \tan A \tan B = -\frac{25}{3}$$

$$\tan(A+B) = \frac{\frac{10}{3}}{1 + \frac{25}{3}} = \frac{5}{14}$$

$$\sin(A+B) = \frac{5}{\sqrt{196+25}} = \frac{5}{\sqrt{221}}$$

$$\left(\frac{25 \times 3}{221} - 10 \times \frac{70}{221} - \frac{25 \times 116}{221} \right) = 25$$